



M/J Comprehensive Science 3

Grade 8 · Course #2002100 · Full Year (Semesters 1 & 2) · Optima Academy Online

Family Course Overview

A one-page guide to the year ahead

Comprehensive Science 3 is the capstone of middle school science: a yearlong journey through **life science, physical science, and space science** that deepens scientific literacy and prepares students for high school. Eighth graders reason from evidence, design their own experiments, and explore matter from the atom up to black holes — finishing with focused review for Florida's statewide science assessment.

THE YEAR AT A GLANCE

Q1 Life Science & the Science of Gravity

~8 weeks

A life-science deep dive and unit test, then a pivot to physical thinking: mass vs. weight and the science of gravity, closing with an end-of-quarter reflection.

Q2 Matter, Atoms & the Elements

~9 weeks

Physical properties, the atom and its history, Bohr models and valence electrons, substance debates, and changes in matter — ending with a physical science test and a first look at space.

Projects: Elements in Action — designing the "City of Matter" — and The Atom Museum, a student-curated exhibit on what an atom is.

Q3 Space Science & the Working Scientist

~9 weeks

Earth and space science, the Sun, and what it really means to be a scientist — students launch their own mini-experiments and investigate common space misconceptions.

Project: Science Fair Mini-Experiments — a multi-week, checkpoint-based experiment students design and run themselves.

Q4 Deep Space, Artemis & the Year in Review

~7 weeks

Black holes and the strange frontiers of science, NASA's Artemis program, then a full-year review, statewide-assessment practice, and the engineering mindset.

A TYPICAL WEEK

- ▶ **Read:** Quick Reference Guides that introduce and anchor each concept
- ▶ **Do:** 1–3 weekly assignments — investigations, debates, museum exhibits, and creative builds
- ▶ **Explore:** VR experiences on Optima's virtual campus (photos & recordings submitted in Canvas)
- ▶ **Check in:** quizzes and unit tests at key milestones each quarter

TIME COMMITMENT

250 minutes / week ≈ 4 hrs 10 min

- ▶ Lessons & reference guides: ~60 min
- ▶ Weekly assignments & investigations: ~120 min
- ▶ VR activities & exploration: ~45 min
- ▶ Quizzes, discussions, review: ~25 min

Weekly minutes come from Optima's master schedule; the per-component split is an estimate. Experiment-checkpoint and project weeks may run longer.

WHAT STUDENTS NEED

- ▶ VR headset for Optima's virtual campus
- ▶ Science notebook, pencils, highlighters
- ▶ Simple household materials for the science fair experiment & projects
- ▶ **On AI:** students learn to use AI thoughtfully — the course includes its own AI-use rubric, guided by the seven Pillars of Virtue

ON-DEMAND (ASYNCHRONOUS)

This course is designed for self-paced learning: students work through the weekly modules on their own schedule with a teacher available for support, feedback, and grading.

LIVE SCHOOL

Live-school students complete the same course in VR every school day, together with a teacher and classmates in real time.

PLEASE NOTE

This overview describes the course as designed. The week-to-week experience may vary from teacher to teacher, as each teacher leans into their own strengths and passions.

Optima Academy Online · Grading: 60% summative (tests, quizzes, projects) · 40% formative (weekly assignments)